# AI for Medicine – Project Report

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## 1. Project Title

A Machine Learning Pipeline for Diabetes Prediction from Clinical Data Using the Pima Indians Dataset

## 2. Problem Statement

Diabetes is a chronic disease that, if not detected early, can lead to serious problems such as heart disease, kidney failure, and nerve damage. If the risk of someone having diabetes is predicted early by using easily measured clinical data (for example glucose, blood pressure, BMI, and age), it can support faster, less expensive diagnosis and reduce long-term health and economic costs. Scientifically, this project is motivated by the need to understand how combinations of easily accessible clinical data (like glucose, blood pressure, BMI, and age) relates to risk of an individual having diabetes and to compare different machine-learning methods to realistically capture this relationship. We use the PIMA Indian dataset to do so. Similar approaches have been studied in the past to predict diabetes using the same dataset.[3]

## 3. Objective of the Study

The specific objective of this project is to build and evaluate a machine-learning pipeline that predicts whether a patient has diabetes or not (binary classification: diabetes (1) vs. no diabetes (0)) using easy to access clinical data from the Pima Indians Diabetes dataset. We aim to compare different models (Logistic Regression, Random Forest, and XGBoost) under the same validation strategy, then select the best-performing model based mainly on ROC-AUC and analyze which clinical features contribute most to the final model's predictions.

## 4. Dataset Description

The dataset used in this project is the *Pima Indians Diabetes Dataset*, originally collected by the US National Institute of Diabetes and Digestive and Kidney Diseases accessible from Kaggle.

### Number of samples

The dataset contains 768 patient records with 8 input features and 1 binary target variable.

### Type and description of data

The data are tabular clinical measurements from adult women of Pima Indian heritage. Features include number of pregnancies, plasma glucose concentration, diastolic blood pressure, triceps skin-fold thickness, two-hour serum insulin, body-mass index (BMI), diabetes pedigree function, and age. Each row represents a single patient.

### Available labels

The target variable (Outcome) is binary:

- 0 → No diabetes (tested negative)
- 1 → Diabetes (tested positive).

### Main Challenges

The dataset shows moderate class imbalance, with 500 non-diabetic and 268 diabetic samples. In addition, some features contain physiologically impossible zero values (e.g., Glucose, Blood Pressure, Skin Thickness, Insulin, BMI), which are treated as missing data during preprocessing and dealt with inside the ML pipeline.

Class distribution for the dataset is shown in figure 1, showing the moderate imbalance.

Figure 3 shows the feature distribution with histograms for each feature. It is notable that features like glucose level and BMI have an important role in the increase of risk of diabetes.

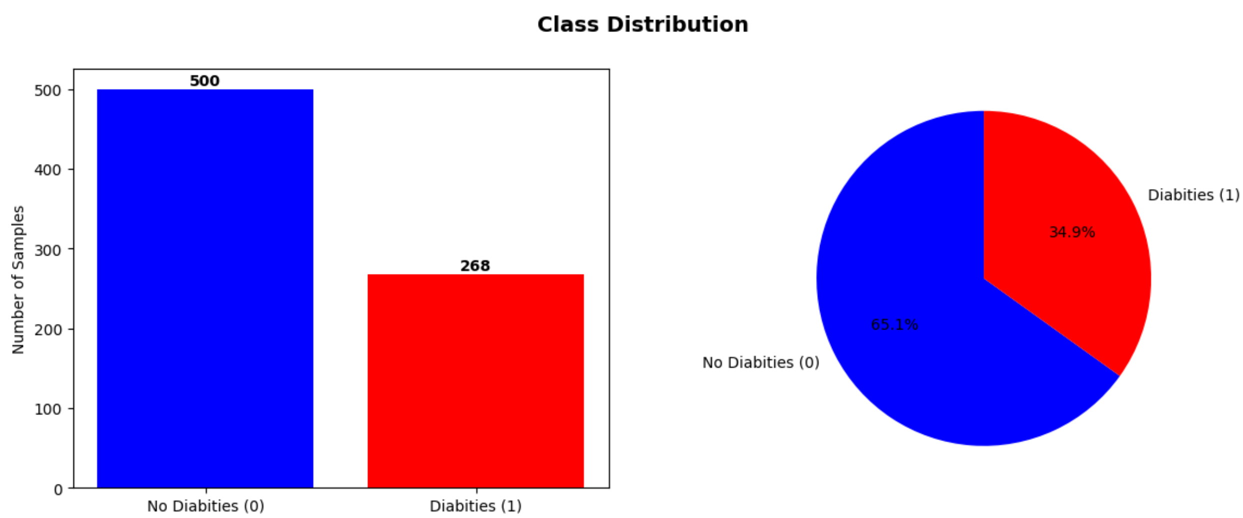

Figure 1: Class Distribution

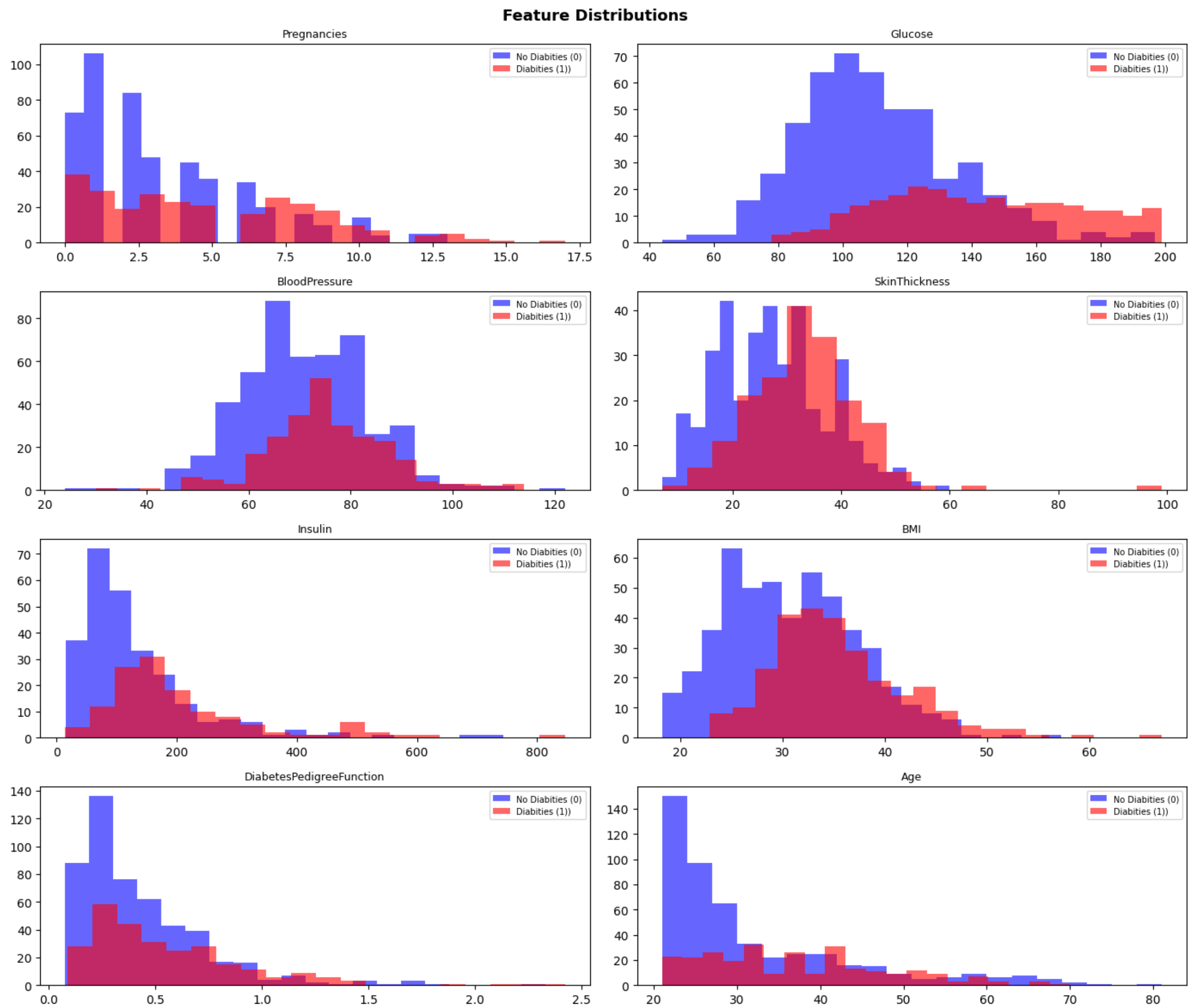


**Figure 2: Feature Distribution**

## 5. Data Preprocessing

The dataset was first loaded from GitHub into a panda Data Frame, and the input features (X) were separated from the binary targets (Y) (Outcome). Then, physiologically impossible zero values in Glucose, Blood Pressure, Skin Thickness, Insulin, and BMI were replaced with missing values (NaN), so they could be treated as missing clinical measurements not valid zeros.

For all models, data cleaning and normalization were implemented inside scikit-learn pipelines: each pipeline uses a SimpleImputer with median strategy to calculate the missing values, followed by StandardScaler to standardize the features before classification.

## 6. Avoiding Data Leakage

In this project, special care was taken to avoid data leakage.

1. All model tunings are done with nested cross-validation instead of tuning directly on the whole dataset. In the inner loops of the CV, each model's hyperparameters are chosen using only the training folds, while the outer loops give an unbiased estimate of performance on test folds that were not used for model design or tuning.
2. Preprocessing step like median imputation of missing values, and feature scaling with StandardScale are implemented inside scikit-learn pipelines. This means that imputation and normalization are fitted only on the training data in each fold. No preprocessing is ever fitted on the full dataset before cross-validation, which would leak information from the validation and test data back into the training data.
3. The same principle is used for model selection: the choice between Logistic Regression, Random Forest, and XGBoost is based on cross-validated metrics, without using the final evaluation plots to adjust hyperparameters or thresholds. The confusion matrix, ROC curve, and feature-importance analyses are produced only after the final model and hyperparameters have been fixed.
4. While evaluating the final model, cross_val_predict is used to generate predictions. Each prediction for a given patient is made by a model that was trained only on other patients, so the data of that patient is never used in the training step that produces its prediction.

## 7. Machine Learning Pipeline

The machine-learning workflow was organized into four main steps: defining the models, validating them, choosing the best one, and then interpreting its results.

**Models compared:**

- Logistic Regression
- Random Forest
- XGBoost

All three models share the same preprocessing steps (imputer + StandardScaler) inside their pipelines to keep the comparison fair and to prevent leakage.

**Validation strategy**

To get a reliable and unbiased estimate of performance on this small, imbalanced dataset, nested stratified cross-validation is used:

- Outer loop: 5-fold StratifiedKFold, providing five folds used only for final performance estimation.

- Inner loop: 5-fold StratifiedKFold inside GridSearchCV, used purely for hyperparameter tuning inside of each outer-loop training set.

**Hyperparameter tuning**

Grid search was used for model hyperparameter tuning. The parameter grids included hyperparameters such as:

- Logistic Regression: regularization strength and class weight
- Random Forest: number of trees, maximum depth, and related tree settings
- XGBoost: number of estimators, learning rate, and depth-related settings.

**Choosing the best model**

Within each outer fold, cross_validate computes several metrics on the held-out data, using roc_auc as the main optimization target and it will report:

- AUC (ROC-AUC) as the primary performance measure.
- F1-score, precision, recall, and accuracy as additional metrics

After comparing the three pipelines the model with the highest mean ROC-AUC will be selected as the final model. Then the best hyperparameters are found in a new inner cross-validation loop for the final model, this model then will be retrained on the entire dataset (all 768 patients) with the full preprocessing pipeline (imputation and scaling). This final, fully trained model is the one used for the subsequent ROC curve, threshold analysis, confusion matrix, and feature-importance discussions in the report.

## 8. Results

### Cross-validated models performances

Using the nested 5×5 stratified cross-validation setup, all three pipelines achieved reasonably performance. The table below shows the achieved metrics.

| Model | Mean AUC | Mean F1 | Mean Precision | Mean Recall | Mean Accuracy |
|---|---|---|---|---|---|
| Logistic Regression | 0.837 | 0.6428 | 0.6852 | 0.6233 | 0.7578 |
| Random Forest | 0.8362 | 0.6312 | 0.712 | 0.57 | 0.7682 |
| XGBoost | 0.8331 | 0.6643 | 0.6988 | 0.6342 | 0.776 |

## Final model and ROC curve

Based on the nested cross-validation summary, logistic regression was selected as the final model because it achieved the highest mean ROC-AUC while maintaining a good balance between F1, precision, recall. After this selection, the hyperparameters were tuned again and the final logistic regression model was then retrained on the entire dataset (all 768 patients).

After training the final model, the ROC curve was computed on predictions using cross_val_predict, to avoid data leakage. The curve is shown in figure3, and the area under the curve is consistent with the nested CV AUC (~0.84). the ROC-Curve confirms that the model can rank diabetic and non-diabetic patients reliably across a range of thresholds.

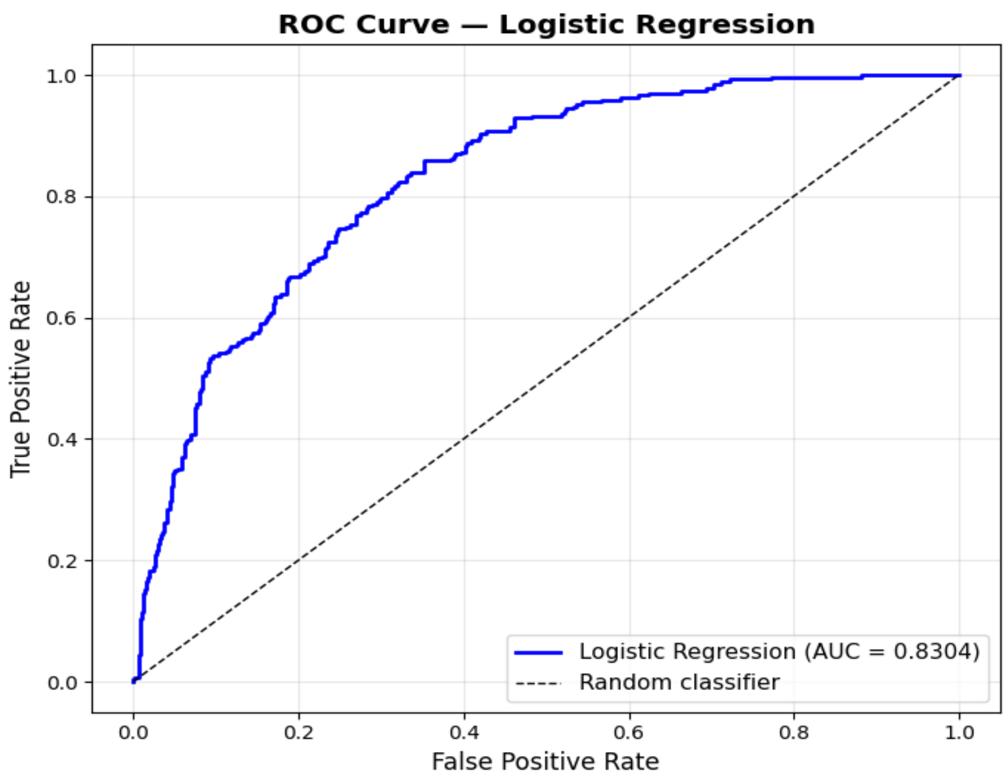


Figure 3: Final Model - ROC-Curve

## Threshold selection and confusion matrix

Because diabetes screening has asymmetric costs (missing a diabetic patient is worse than flagging a non-diabetic one), instead of using the default 0.5 the threshold was tuned with the following approach. To avoid missing true diabetic patients as much as possible, the search was restricted to thresholds between 0.1 and 0.5, which encourages higher recall. Within these thresholds the program finds the one that maximizes the f1 score.

At the selected threshold, the confusion matrix (figure 4) shows that the model correctly classifies most non-diabetic patients. False negatives are lower than with a simple 0.5 threshold, at the price of some extra false positives, which is acceptable for a screening tool because positive patients can be sent for more accurate follow-up tests.

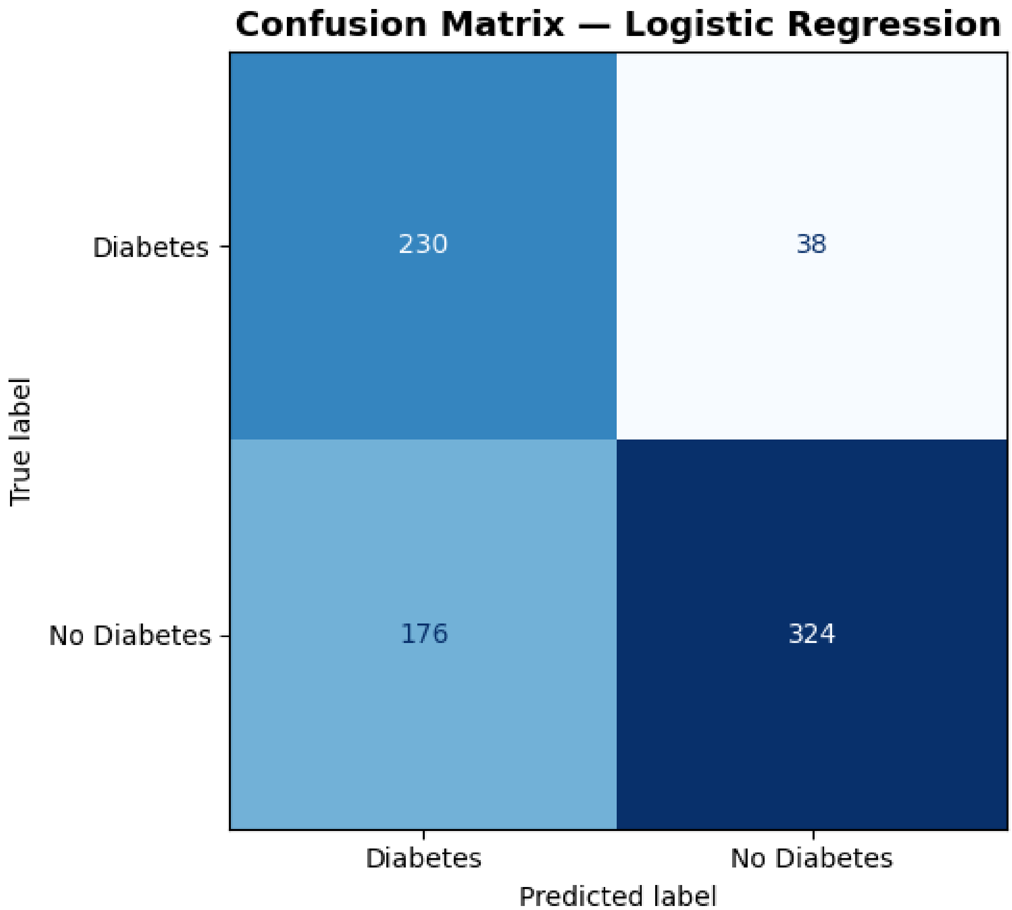


Figure 4: Final Model- Confusion matrix

The table below shows the achieved performance for the final model.

| Parameter | Threshold | F1 | Precision | Recall | Accuracy |
| --- | --- | --- | --- | --- | --- |
| Value | 0.31 | 0.6825 | 0.5665 | 0.8582 | 0.7214 |

## Feature importance and clinical interpretation

To better understand what the model has learned, the final logistic regression coefficients were examined and plotted after standardization. Features such as Glucose level, BMI, and number of pregnancies show the largest positive coefficients, meaning that higher values of these variables are strongly associated with an increased predicted risk of diabetes. In contrast, features like lower younger age have lower coefficients, indicating that the lower they the lower the risk of diabetes.

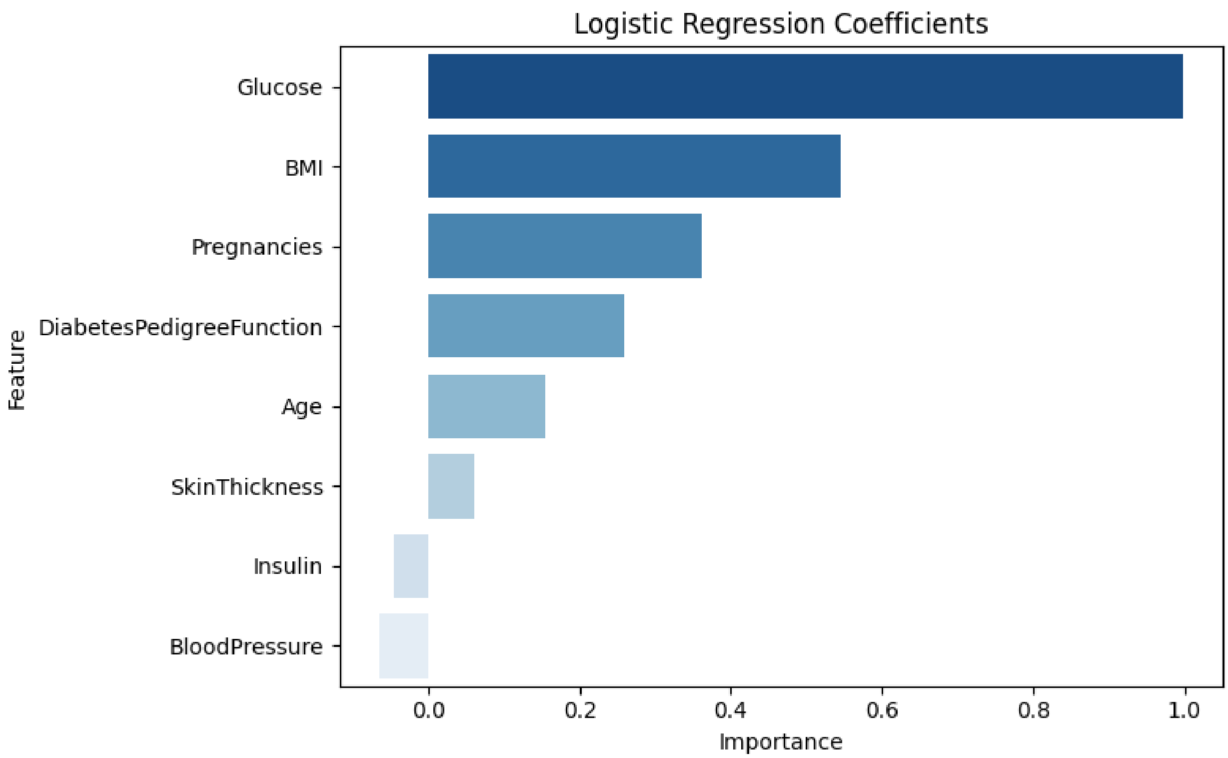


**Figure 5: Final Model - Feature Importance**

## 9. Discussion

Results show that the final logistic regression model can work as a reasonable screening tool for diabetes on the Pima Indians dataset. The model reaches a ROC-AUC of about 0.83–0.84 and an F1-score around 0.65, which means it can meaningfully separate diabetic from non-diabetic patients. However, shown by the confusion matrix, when choosing the threshold, I explicitly sacrificed precision to gain recall, the model catches more true diabetic patients (fewer false negatives) at the cost of more false alarms, which is acceptable in a screening context where flagged patients can be sent for further testing. Additionally, the feature distribution plots (Figure 2) already show that diabetic and non-diabetic patients differ clearly in variables such as glucose, BMI etc., and the feature-importance plot (figure 5) for the final model confirms this by assigning the largest positive coefficients to exactly these features. Together, these two figures suggest that the model is not relying on patterns, but mainly on clinically risk factors that are visible in the raw data.

Despite these positive results, there are clear limitations. The dataset is small (768 patients) and moderately imbalanced, so the performance numbers could change with new data. Additionally, all subjects are adult women of Pima Indian heritage, so the model may not work equally well for other populations without retraining. Moreover, the model only uses a few clinical measurements and does not include information such as medication, or lifestyle factors, which likely limits its maximum achievable performance. For these reasons, the model should be seen as a proof of concept for low-cost diabetes screening, not as a ready-to-use clinical tool, and future work should focus on larger, more diverse datasets and prospective evaluation in real practice.

## 10. Conclusions and Future Work

In this project I built a complete pipeline for diabetes prediction on the Pima Indians dataset using simple, interpretable models. Logistic regression, evaluated with nested stratified cross-validation, achieved the best ROC-AUC (~0.84), across folds, outperforming Random Forest and XGBoost under the same conditions. By tuning the threshold, the final model was created. it intentionally accepts more false positives to reduce false negatives and capture more true diabetic patients.

In the future, the model should be tested on more data from more diverse people, not only adult Pima Indian women, to see if the results hold true in other populations and settings. Collecting extra information such as more lab tests, follow-up measurements, and basic lifestyle data could help the model make more accurate predictions. Finally, it would be useful to try the model in a realistic clinical environment and adjust the threshold together with clinicians, to ensure that the balance between missed cases and false alarms matches real-world needs.

## 11. Ethics and Data Privacy

The project uses the Pima Indians Diabetes dataset, which is publicly available for academic and research use (e.g., via Kaggle and UCI) and does not contain directly identifiable personal information. The data were used only for methodological experimentation and model development, with no attempt to re-identify individuals or link the data to external sources.

## 12. Code and Reproducibility

The main code for this project is contained in the Jupyter/Colab notebook **AI_Project_PIMA_.ipynb** . To reproduce the results, a standard Python 3 environment with the following libraries is required: NumPy, pandas, scikit-learn, XGBoost, matplotlib, and seaborn. Running the notebook from top to bottom will automatically download the dataset from GitHub, perform all preprocessing, train and evaluate the models with nested cross-validation, and generate the figures and tables reported in this document; the notebook, the written report, and the saved result files are accessible within the project github repository linked below.
*https://github.com/Reza-Jokarian/AIForMedicine-FinalProject*

## 13. References

[1] National Institute of Diabetes and Digestive and Kidney Diseases. (1990). *Pima Indians Diabetes Database* [Data set]. Donated to the UCI Machine Learning Repository by V. Sigillito, Johns Hopkins University Applied Physics Laboratory

[2] UCI Machine Learning Repository. (n.d.). *Pima Indians Diabetes Database.* University of California, Irvine. Retrieved for academic use via Kaggle and UCI.

[3] Nassiwa, F., & Zeng, J. (2016). Evaluating traditional machine learning models for predicting diabetes onset using the Pima Indians dataset. *Annals of Medical and Health Sciences Research.*

[4] Perplexity AI. (2026). *Perplexity, powered by GPT-5.1, assistance in coding and ML concepts*